

Noah Hickman

(469)-401-0911 · noahh3@me.com · [linkedin.com/in/noahh23](https://www.linkedin.com/in/noahh23) · Portfolio: noahh-2305.github.io

EDUCATION

B.S. Mechanical Engineering – The University of Texas at Austin

May 2027

• GPA: 3.84; Relevant Coursework: Robot Mechanism Design, MATLAB

SKILLS

Programming & Data: Python (pandas, NumPy, matplotlib, scikit-learn, psycopg2), MATLAB, SQL, Docker/Portainer, Git

Simulation & Testing: ANSYS Mechanical (FEA), ATI Vision, Vector CANalyzer, Simulink, Hondata/KTuner

CAD & Manufacturing: SolidWorks, Fusion 360, TIG Welding, 3D Printing, GD&T, Lathe and Manual Mill, Soldering

General: Microsoft Office, Confluence, Jira, Agile, U.S. citizen with no work restrictions, trained in ITAR compliance

WORK EXPERIENCE

Product Testing Mechatronics (PTM) Intern – Daimler Trucks North America

May 2025 – Aug. 2025

- Programmed and deployed a standalone Python script package with custom machine learning and localized semantic models to transcribe voice trigger logs, detect optical acoustic warnings (OAWs), and automate test data tagging
- Merged Python tools with the PTM data platform and SQL backend, streamlining analysis across multiple testing teams
- Executed PVE vehicle feature testing (ADAS/Brakes) and data acquisition during Pacific Northwest Over-the-Road tests

Powertrain Integration Engineering Co-Op – Volvo Group

Jan. 2025 – May 2025

- Performed root cause analysis on diesel powertrain faults and sensor logic using MATLAB and Simulink documentation
- Integrated chassis and powertrain hardware, harnesses, and sensors to validate CARB-compliant engine packages
- Extracted CAN data from vehicle ECMs using ATI Vision and CANalyzer to troubleshoot 56+ engine and exhaust parameters
- Coded custom Python tools for statistical analysis of powertrain data, reducing manual troubleshooting time by 50%

TECHNICAL ACTIVITIES AND PROJECTS

Combustion Engine Calibration Lead – Longhorn Racing Combustion (Formula SAE)

May 2025 – Present

- Spearheading calibration and sensor integration on Honda CBR600RR engine using a Haltech Elite 1500 ECU for event-specific performance tuning across 4 testing events
- Leading research and testing of E85 fueling system integration for the next-generation car

Longhorn Racing Solar (Formula Sun Grand Prix, American Solar Challenge)

Aug. 2023 – May 2025

Roles: **Mechanical Systems Mentor · Body Lead · Frame Sublead · Member**

- Served as an advisor for design reviews and member training, mentoring over 45 mechanical team members
- Led the 15-member Frame/Ergonomics subsystems, organized meetings, and created a high-level design roadmap
- Reduced ANSYS simulation time by 80% by generating MATLAB scripts to automate stiffness calculations
- Launched team-wide research on structural composites, testing alternatives to conventional frame structures
- Utilized Ansys Mechanical FEA to validate and complete chassis design for legal road-use based on ASC regulations
- TIG-welded 114-tube space frame assembly with .001" tolerance mounting tabs, using custom lathed aluminum pins, laser cut wood paneling, and 3D-printed tab mount welding fixtures
- Modeled tubular frame roll-cage structure and cross-bracing in SolidWorks; supported suspension part manufacturing

Propulsion Subsystem Member – Longhorn Rocketry Association (Spaceport America)

Feb. 2024 – Jul. 2024

- Designed a converging-diverging graphite nozzle and an impinging injector for the Taurus II hybrid engine using SolidWorks
- Conducted trade studies between 10+ materials for optimal performance across nozzle O-rings and O-ring lubricant

Flight-Weight Fluids System Member – Texas Rocket Engineering Lab (Spaceport America)

Aug. 2023 – Aug. 2024

- Compiled valve/fitting specs from P&ID schematics to meet high-pressure fluid system requirements
- Operated a LabQuest DAQ to characterize pressurant flow on an orifice flow bench at Firefly Aerospace test facility

Personal Project Car, 1997 Acura Integra

Ongoing

- Planned and integrated multiple fueling, cooling, and drivetrain upgrades with multiple upcoming installs
- Street-tuned via Hondata S300 board with goal of designing and building a custom supercharger system for B18B1 engine

Hobbies: Project car building, record/vinyl collecting, and solo traveling; Member of UT ASME